PREPARE

Advancing Crop Price Prediction and Farmer Empowerment

The Team

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Problems

1: Farmer's Price Blindness

Small and marginal farmers lack timely, accurate price forecasts and struggle to interpret market dynamics.

2: Ineffective Government Interventions

Governments and regulators lack reliable multi-day price prediction tools, making it difficult to set fair MSPs or plan timely market interventions.

The Engine

- Input Data: Historical Price, Geographical Location, Weather
- Data Pre-processing: Outlier Removal, Normalization
- Modeling Core: Define and build opon the model chosen
- Output: Accurate Price Predictions

The Chassis

- Prediction Service: Refined model is exposed via an API or server.
- Farmer Application: A mobile application for practical deployment.
- End-User Value: Empowers farmers with timely and accurate price forecasts

Schematic

Novelty

- Capturing the interdependent nature of different markets for food pricing.
- There are **no current consumer facing solutions** that provide price predictions.

Real World Application

This helps farmers to decide when to harvest crops and where* to sell those crops for optimal returns. The format of the application also makes it accessible for farmers and does not have a significant learning curve. This can also be applied on the policy side to identify when interventions are needed from the government.

Model Development and Results

We experimented with and developed models for 10 day price prediction across all APMC mandis in India and settled on a GAT-LSTM model with satisfactory metrics for prediction.

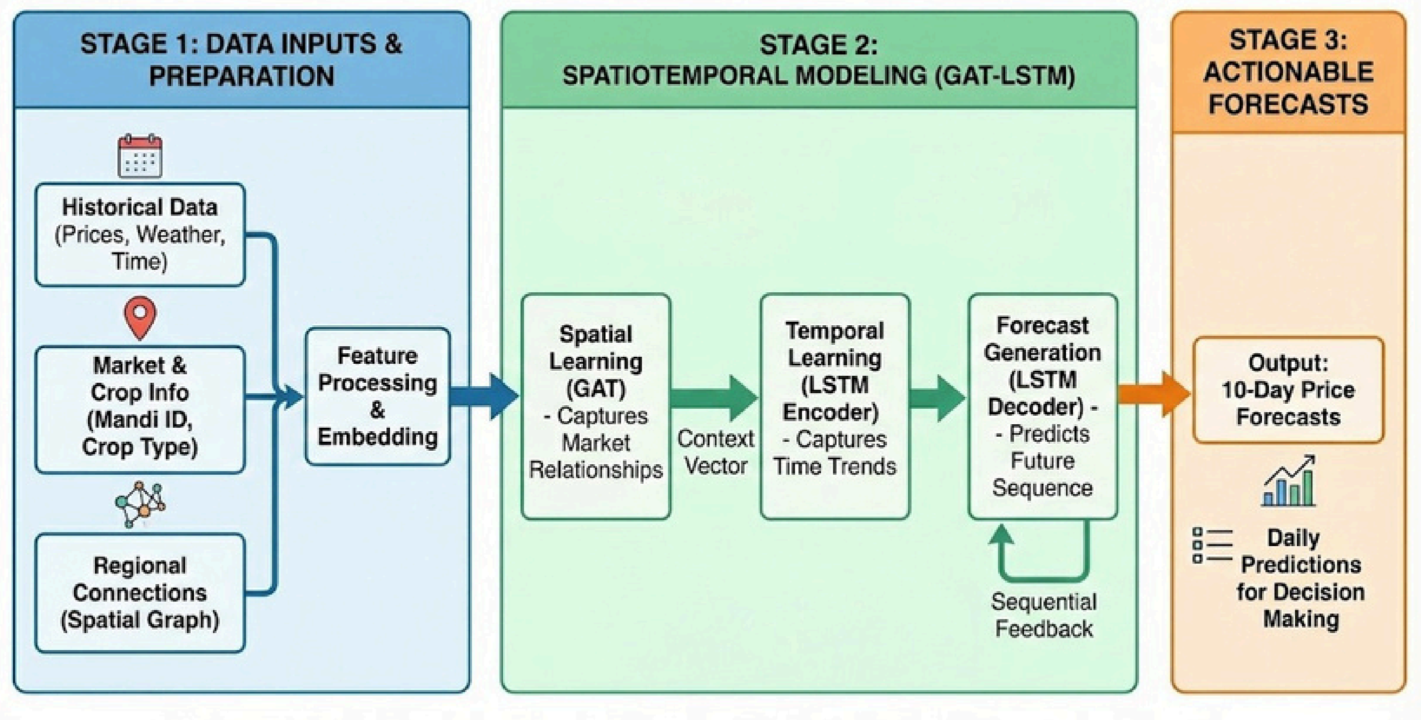

Figure 1: Application Flow of the GAT-LSTM Price Forecasting System

Metrics

Day	RMSE (₹/100kg)	MAE (₹/100kg)
1	252.59	177.53
2	288.60	198.90
3	314.08	217.34
4	333.61	231.02
5	348.95	241.72
6	361.51	250.93
7	371.87	258.71
8	381.00	265.79
9	388.82	272.09
10	395.71	277.85

Metrics

Day	RMSE (₹/100kg)	MAE (₹/100kg)
1	146.51	76.00
2	158.40	75.70
3	167.15	79.79
4	174.33	81.23
5	179.95	82.08
6	183.63	82.74
7	185.85	83.39
8	187.44	84.21
9	188.95	85.29
10	190.45	86.53

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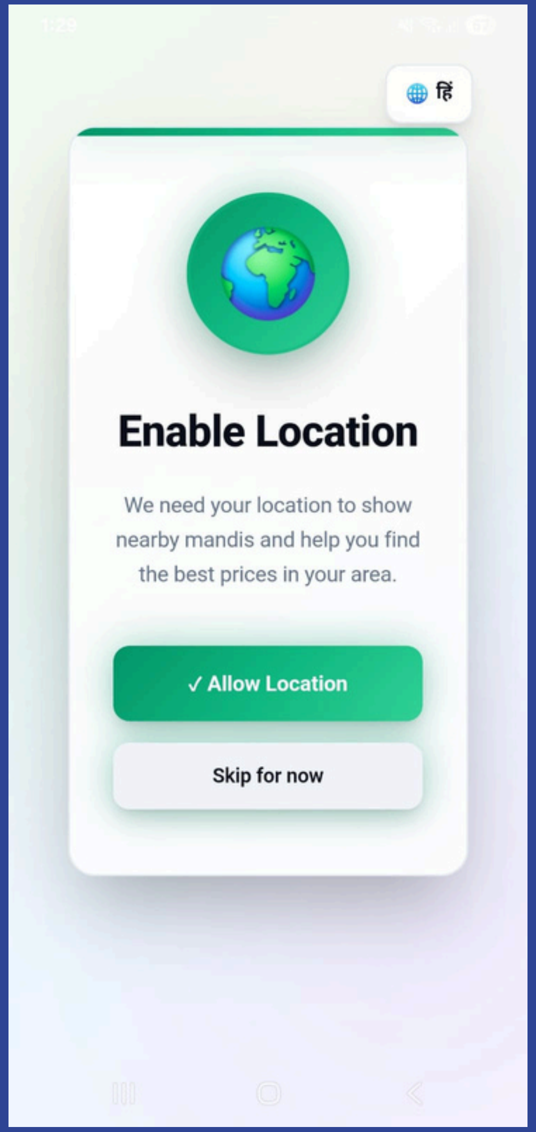
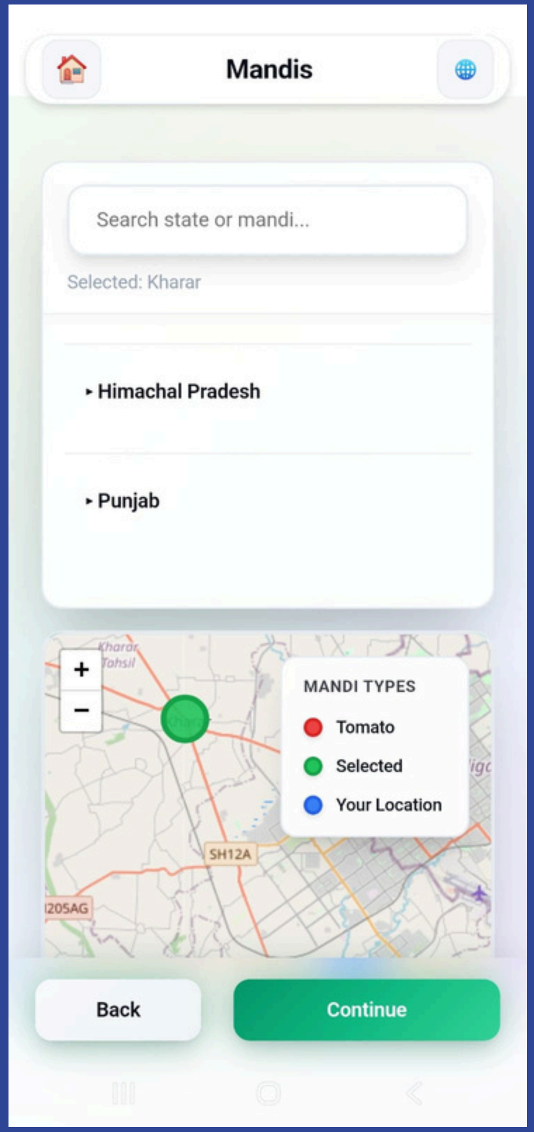
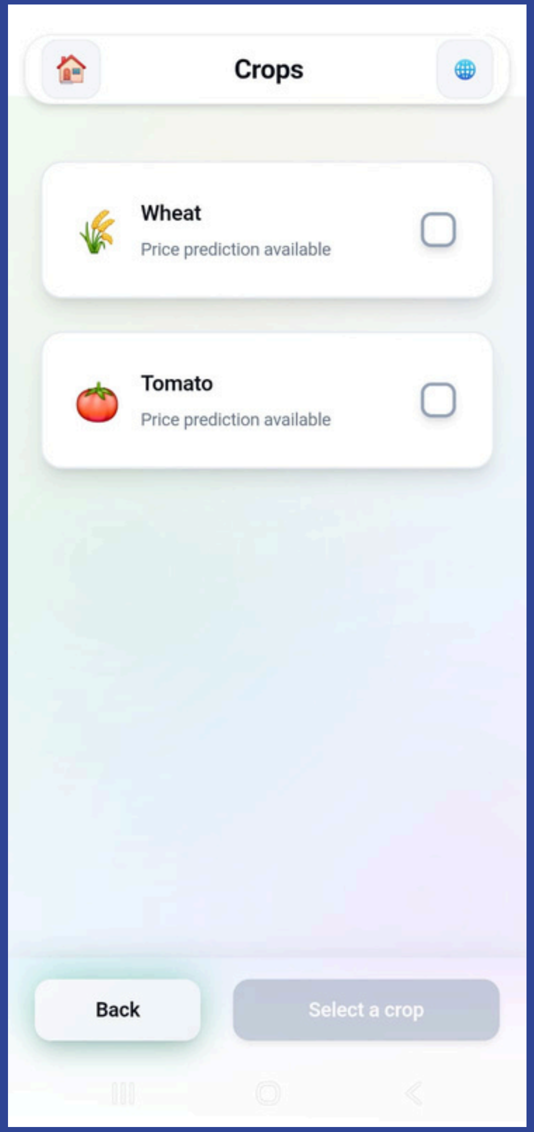
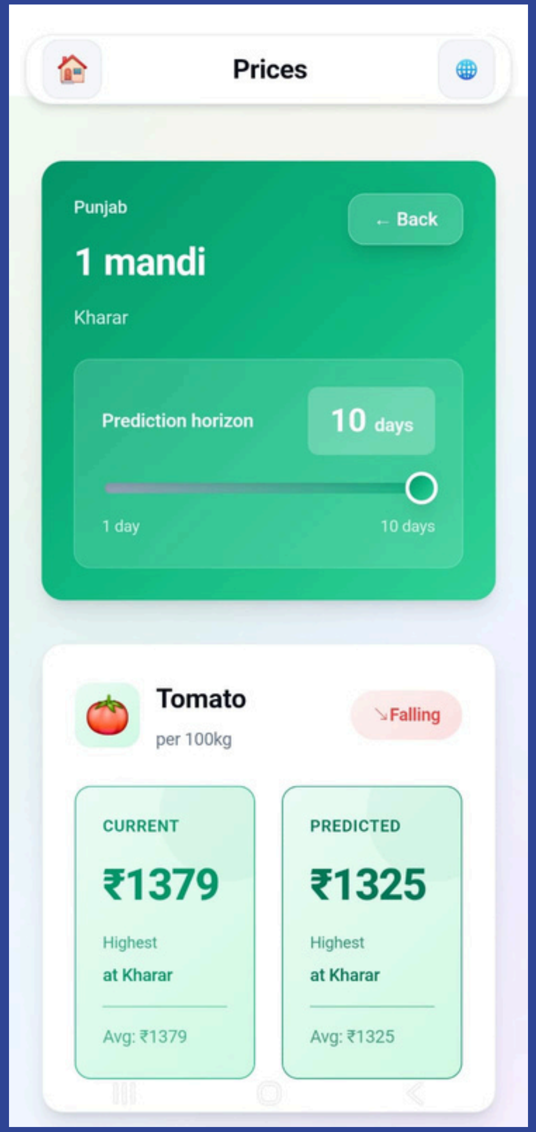
Mobile Interface Development









We developed a mobile interface with features such as location detection, day-based price estimation, and multiple languages and linked our models to the interface using FastAPI.

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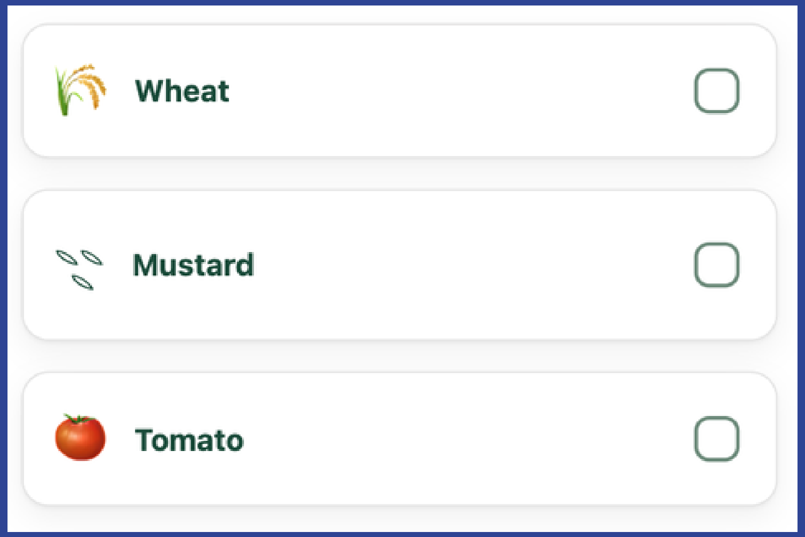
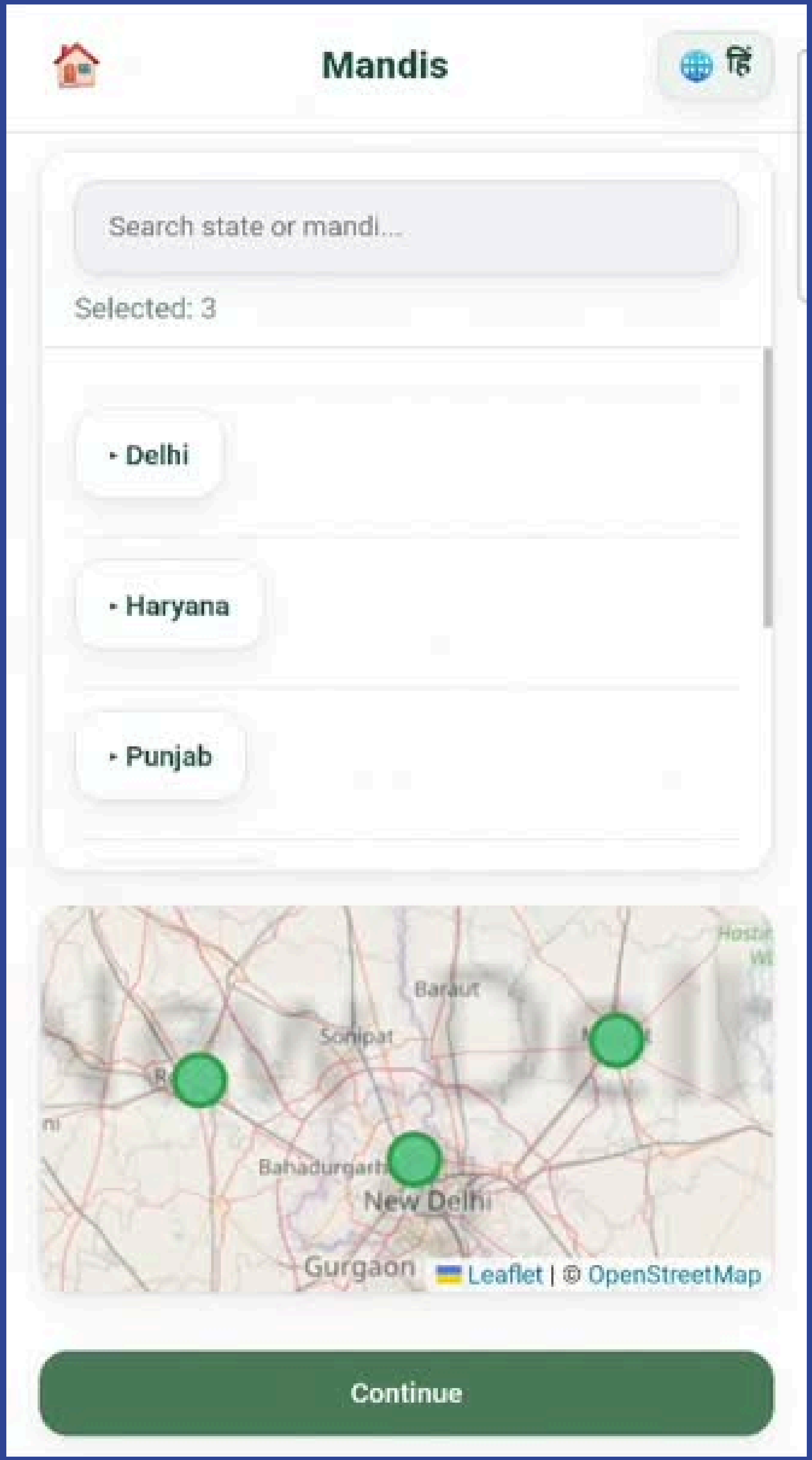

Wheat
Mustard
Tomato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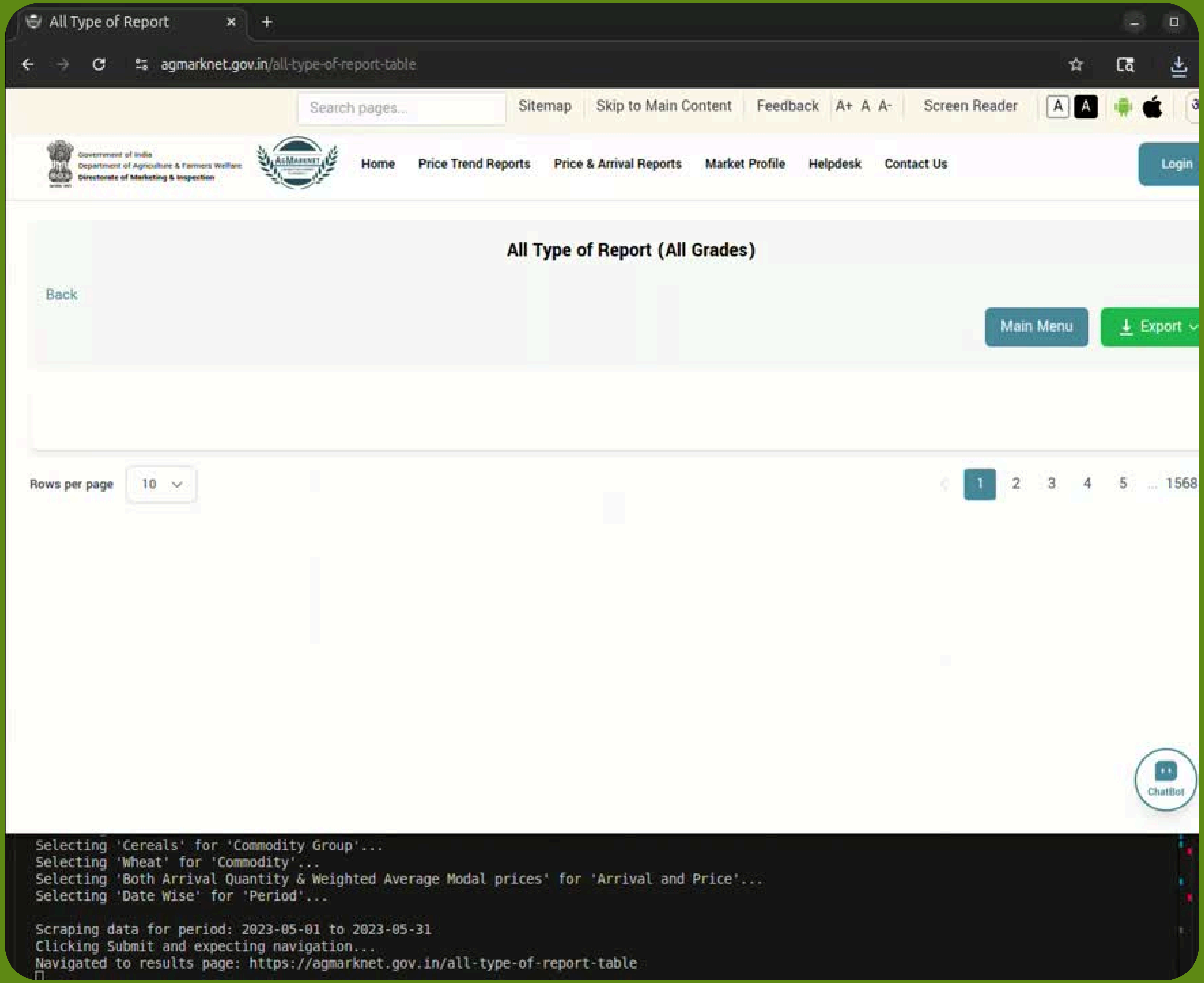
Data Acquisition

We extracted daily agricultural records from the Agmarknet portal (Jan 2023 – Jan 2026) using custom Python and Playwright scraping scripts. To accurately expose missing data, we expanded the raw extractions into a strict date-by-mandi calendar grid. This ensured every mandi had a dedicated row per day, representing missing arrival quantities and modal prices as explicit null values.

सत्यमेव जयते

Government of India
Department of Agriculture & Farmers Welfare
Directorate of Marketing & Inspection

AGMARKNET
CONNECTING FARMER TO MARKETS



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1 2 3 4 5 ... 1568
ChatBot
Selecting 'Cereals' for 'Commodity Group'...
Selecting 'Wheat' for 'Commodity'...
Selecting 'Both Arrival Quantity & Weighted Average Modal prices' for 'Arrival and Price'...
Selecting 'Date Wise' for 'Period'...
Scraping data for period: 2023-05-01 to 2023-05-31
Clicking Submit and expecting navigation...
Navigated to results page: https://agmarknet.gov.in/all-type-of-report-table

EDA

Top 15 mandis with lowest missing % (most complete):

State	Market	Days in Range	Present	Missing	% Missing
Madhya Pradesh	Badamalhera APMC	1,065	741	324	30.4%
Uttar Pradesh	Mugrabaadshahpur APMC	1,065	741	324	30.4%
West Bengal	Khatra APMC	1,065	749	316	29.7%
Madhya Pradesh	Dhar APMC	1,065	764	301	28.3%
Uttar Pradesh	Tulsipur APMC	1,065	770	295	27.7%
Madhya Pradesh	Umariya APMC	1,065	772	293	27.5%
Uttar Pradesh	Jayas APMC	1,065	772	293	27.5%

We initially targeted five crops, but EDA on the expanded grid revealed severe reporting sparsity. Sugarcane was dropped due to an extreme 98.6% missing data rate. Remaining crops also showed massive gaps; for example, Wheat was missing 75.6% overall and 90.87% on Sundays. We also found hundreds of mandis with uninterrupted missing streaks lasting over a year, prompting us to finalize our scope to Onion, Potato, Tomato, and Wheat.

Location and Weather Data

We mapped exact latitude and longitude coordinates to every mandi and integrated the Copernicus API to extract historical, daily weather features for each location. This transformed our basic price ledger into a multi-dimensional dataset capable of capturing the real-world variables driving agricultural markets.

Weather & Location Data

Column	Description	Non-null
latitude	Latitude (°)	1,650,576 (100.0%)
longitude	Longitude (°)	1,650,576 (100.0%)
t	Temperature (°C)	1,650,576 (100.0%)
tp	Total Precipitation (m)	1,650,576 (100.0%)
ssr	Surface Solar Radiation (MJ/m²)	1,650,576 (100.0%)
r	Relative Humidity (%)	1,650,576 (100.0%)

Price Imputation

To resolve massive data gaps, we researched and came up with ten imputation strategies (ranging from simple baselines to Matrix Factorization) that we benchmarked at ≥0%, ≥50%, and ≥75% data presence thresholds. Through this, we identified a day of the week based ratio to be the best performing strategy for mandis with atleast 50% data and so that is what we went with.

Ranking — All Mandis (5,806 mandis total) — Modal_Price

Rank	Method	RMSE (mean)	MAE (mean)	MAPE (%)	Coverage (%)	Time (s)
1	8_Spline_Pipeline	396.32	105.01	5.72	99.7	105.3
2	2_Rolling_Mean	436.22	112.78	7.75	93.14	3.36
3	7_DOW_Ratio	449.05	113.41	6.91	88.98	22.28
4	10_SVD_Matrix	476.94	157.89	9.39	88.3	459.68
5	9_Random_Forest	477.01	144.01	8.19	100.0	22.22
6	1_Capped_FFill	600.66	104.03	5.13	89.1	2.22
7	6_FFill_Decay	600.74	126.67	6.96	99.26	44.12
8	3_State_Median	765.26	272.81	14.66	87.6	0.78
9	4_Seasonal	921.38	410.24	28.28	87.76	0.82
10	5_Global_Median	1226.85	694.45	43.69	100.0	0.24

Price Imputation

Ranking — ≥50% (2,645 mandis total) — Modal_Price

Rank	Method	RMSE (mean)	MAE (mean)	MAPE (%)	Coverage (%)	Time (s)
1	7_DOW_Ratio	259.58	120.58	5.78	99.7	8.2
2	2_Rolling_Mean	262.53	122.54	5.85	99.98	1.88
3	9_Random_Forest	298.68	143.01	6.88	100.0	14.35
4	1_Capped_FFill	306.27	114.15	5.44	99.05	0.95
5	8_Spline_Pipeline	315.9	122.93	6.02	100.0	74.95
6	6_FFill_Decay	328.31	133.9	6.56	100.0	9.7
7	10_SVD_Matrix	398.73	169.69	8.45	100.0	83.12
8	4_Seasonal	762.14	468.0	23.77	99.25	0.3
9	3_State_Median	787.07	316.48	15.54	99.25	0.3
10	5_Global_Median	1272.68	793.32	42.23	100.0	0.1

Ranking — ≥75% (1,066 mandis total) — Modal_Price

Rank	Method	RMSE (mean)	MAE (mean)	MAPE (%)	Coverage (%)	Time (s)
1	2_Rolling_Mean	202.32	96.75	4.82	100.0	0.72
2	7_DOW_Ratio	203.38	98.7	4.97	99.93	2.17
3	6_FFill_Decay	222.32	99.0	5.22	100.0	2.25
4	9_Random_Forest	226.58	111.14	5.6	100.0	4.92
5	1_Capped_FFill	227.07	85.07	4.31	99.58	0.35
6	8_Spline_Pipeline	228.27	90.97	4.76	100.0	25.12
7	10_SVD_Matrix	271.18	139.98	7.48	100.0	14.32
8	3_State_Median	437.27	251.05	14.08	98.3	0.08
9	4_Seasonal	617.85	383.54	20.13	99.5	0.1
10	5_Global_Median	1063.42	707.14	36.68	100.0	0.02

Final Data

January 2023 to December 2025 (3 years)

Final retained mandi count:

- Onion: 1474 → 764
- Potato: 1388 → 713
- Tomato: 1506 → 693
- Wheat: 1421 → 475

Per-crop retained rows:

- Onion: 470,964
- Potato: 459,012
- Tomato: 492,870
- Wheat: 295,535

The main concern was data sparsity within the AGMarknet platform and data collection at a per-mandi level, resulting in the need to drop mandis and impute prices.

Experiments: Non-Graph

We first experimented with non-graph methods to see how they perform.

Explicit baselines (previous-day price, 7-day rolling mean, 28-day rolling mean)

- Tried first to establish how hard the task already is without learning

HistGB, XGBoost, LightGBM

- Tried as a fast non-deep benchmark and sanity check

Anchored HistGB

- Predict a correction around a baseline such as current price or rolling mean by subtracting this from all prices.

Cross-crop HistGB

- Add one paired crop as an input feature
- Tried to test whether related crops act as leading indicators

Density-bucket / volume-cluster variants

- Separate mandis by graph density and output volume (enforced by new feature with labels)
- Tried to test whether sparse and dense markets need different behavior

Experiments: Graph + Sequence

Neighbor-blend and graph-augmented HistGB
- Add neighborhood information without full deep graph learning
- Tried to isolate whether graph information helps even in simpler models

TCN, LSTM
- Temporal sequence model over recent windows
- Tried because some crops may benefit more from temporal pattern extraction than explicit graph structure

GraphWaveNet+
- Adaptive deep graph model from spatiotemporal forecasting literature
- Tried as a strong graph benchmark

Self-Adaptive Graph Mixture of Models (SAGMM)
- Tried based on recent literature

GAT-GRU
- Graph attention encoder with sequencing
- Tried to jointly model inter-mandi influence and temporal effect

We then moved on to graph and sequence based methods to see how they perform.

Ablation Factors:

Target: delta_current, delta_roll7, delta_roll28

Graph mode: full graph, temporal-only, geo-only, shuffled graph

Radius threshold: 75 km, 150 km, 300 km, plus finer onion refinement

Edges and Nodes: edge-weight presets and k neighbors

Ablation Inferences

- Subtracting a rolling mean improved 15-day forecasting, with delta_roll28 being the best for onion, potato, and tomato.
- Graph structure did improve 15-day performance when compared against the best non-graph models, however the difference between graph structures (full vs partial) was not significant.
- Radius thresholding mainly mattered for tomato and onion.
- Density and volume based variants did improve over weaker simple baselines in some cases, but neither became the best final non-graph approach, so they were informative rather than decisive.
- Cross-crop correlation was present, especially onion-wheat and potato-onion, but cross-crop features only gave modest gains and did not change the final 15-day winners.

Crop	Paired crop	Corr.
Onion	Wheat	0.769
Potato	Onion	0.682
Tomato	Potato	0.244

Crop	Best Radius (km)
Onion	Consistent at 25-75
Potato	Nearly flat across 75-300 km
Tomato	75
Wheat	Full graph (no radius)

One Day Results

One day results were strong throughout, even compared to baselines.

Crop	Baseline (prev pred)	Model R^2	Best Performing Model
Onion	0.92	0.95	Tuned TCN
Potato	0.95	0.97	Linear Reg
Tomato	0.84	0.92	Tuned TCN
Wheat	0.69	0.79	Tuned LightGBM

Fifteenth Day Results

Fifteenth day results were dominated by graph-based models as opposed to non-graph based ones. We compared them to predicting the rolling mean (at different intervals).

Crop	Baseline (rolling mean)	Model R^2	Best Performing Model
Onion	0.78	0.81	GAT-GRU (75km)
Potato	0.93	0.94	GAT-GRU (300km)
Tomato	0.37	0.39	GAT-GRU (75km)
Wheat	0.66	0.71	GAT-GRU (Full)

Graph vs Non-Graph Results

The best performing graph models tended to outperform non-graph models at the fifteenth day, while non-graph models performed better at one-day predictions.

Crop	Best Non-Graph*	Best Graph	Gain
Onion	0.71	0.81	+0.10
Potato	0.93	0.94	+0.01
Tomato	0.27	0.39	+0.12
Wheat	0.66	0.71	+0.05

App Integration

We then integrated the best performing models into the app. Instead of predicting prices at each call, we chose to predict prices for a day and index them, and then called the indexed results.

/gosain idhar bhideo dal do

Reflection

Takeaways

- Data work was as important as model work.
- Iterative development and creativity/ideation was extremely helpful in figuring out what to do next.
- Graph models and anchoring helped significantly improve the results.
- Useful signals (such as cross crop correlation and mandi production/density grouping) can exist without always becoming decisive gains.

Limitations

- Structurally mising data remains a bottleneck.
- Imputation does not neccesarily reflect the actual prices.
- Tomato suffered from a huge data issue (we identified this by running the same model architecture on other tomato data - 10 years and that worked better).
- We didn’t know the root cause for a lot of the findings/improvements since machine/deep learning treats a lot of the work as black boxes, so there was a degree of speculation involved.

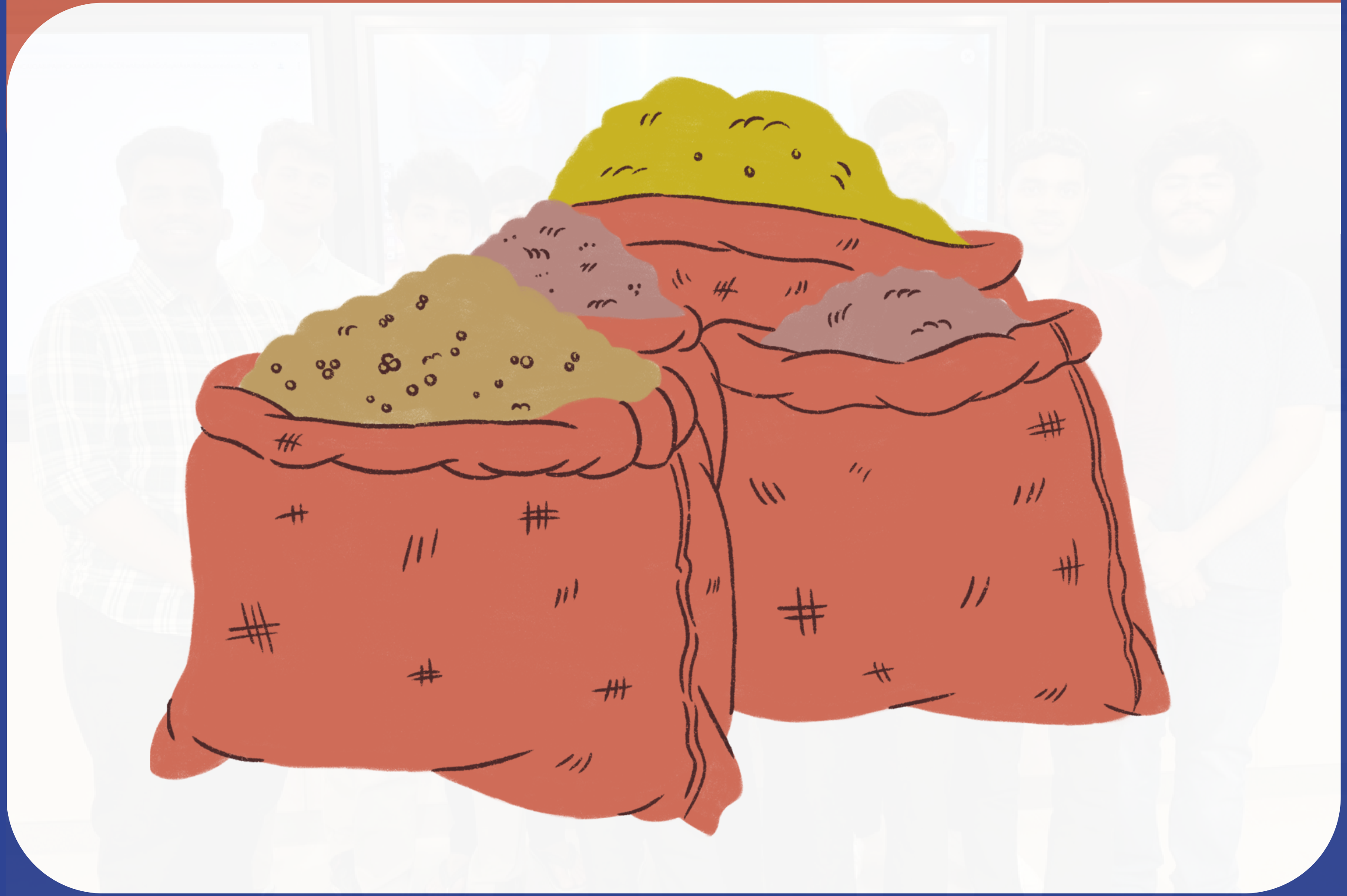
Future Work

Automated Infrastructure to Extend to Other Crops

Game Theoretical Study

Explainable AI

thank



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